MachinePulse

Phase 1 implementation report - reproducible industrial time-series anomaly detection

Implementation overview

MachinePulse uses the UCI MetroPT-3 air-compressor sensor dataset to compare a class-balanced Logistic Regression baseline with a PyTorch LSTM Autoencoder. Phase 1 delivers a fully local and reproducible ML pipeline; deployment, APIs, containers, monitoring, and cloud services are intentionally deferred.

Repository and tracking links

GitHub: github.com/RaoHaris535/MachinePulse-Project

DagsHub: dagshub.com/RaoHaris535/MachinePulse-Project

MLflow: remote machinepulse-phase1 experiment

Phase 1 completion

Data	Official-source ingestion, schema/null/range/time validation, DVC remote
Systems	Logistic Regression baseline and LSTM Autoencoder main model
Tracking	Remote MLflow parameters, metrics, artifacts, and registered Staging model
Pipeline	Six named DVC stages reproducible with one dvc repro command
Quality	17 automated tests, Ruff linting, deterministic seeds, no raw data in Git

Held-out test results

Model	Test F1	ROC-AUC	Avg. precision
Logistic Regression	0.0000	0.0249	0.0023
LSTM Autoencoder	0.0110	0.9825	0.2787

The main model sharply improves threshold-independent anomaly ranking. The low fixed-threshold F1 is retained as an honest limitation caused by temporal distribution shift between failure periods; the held-out test set was not used for threshold tuning.

Data pipeline and versioning

PASSED	1,516,948	0
data validation	raw sensor rows	nulls or duplicate times

Ingestion and validation

The ingest stage streams the official UCI ZIP, extracts only the configured MetroPT-3 CSV, and records SHA-256 hashes in a source manifest. Validation scans the 1.5 million rows in chunks and checks the required schema, row-count bounds, nulls, numeric conversion, finite values, physical ranges, binary domains, timestamp parsing, ordering, and duplicates. Any failed check exits non-zero and blocks preprocessing.

Preprocessing and sequence generation

Raw 10-second readings are aggregated to 5-minute intervals. Published failure windows create active-failure labels and a one-hour pre-failure warning target. Chronological train/validation/test splits isolate later incidents. Scaling is fit only on normal training rows, and the LSTM receives 12-step (one-hour) sequences that never cross missing-time gaps.



Meaningful DVC outputs

Raw and processed data, model bundles, figures, and JSON metrics are cached by DVC and synchronized to the DagsHub S3-compatible remote. Git stores only code, configuration, dvc.yaml, and dvc.lock.

Reproducibility controls

All stage dependencies and outputs are explicit. Seeds, split boundaries, failure windows, model hyperparameters, and early-stopping settings are versioned in YAML. The validation report is a dependency of preprocessing, making validation a true training gate.

Remote status

After the final run, dvc push reported everything up to date and dvc status -c confirmed that the local cache and remote origin were synchronized.

Figure 1. Live DVC pipeline output after validation was made a training gate.

Models and remote experiment tracking

The Logistic Regression baseline uses class balancing and validation-F1 threshold selection on tabular intervals. The LSTM Autoencoder is trained only on normal training sequences, uses reconstruction MSE as its anomaly score, and restores the best validation-normal checkpoint through early stopping.

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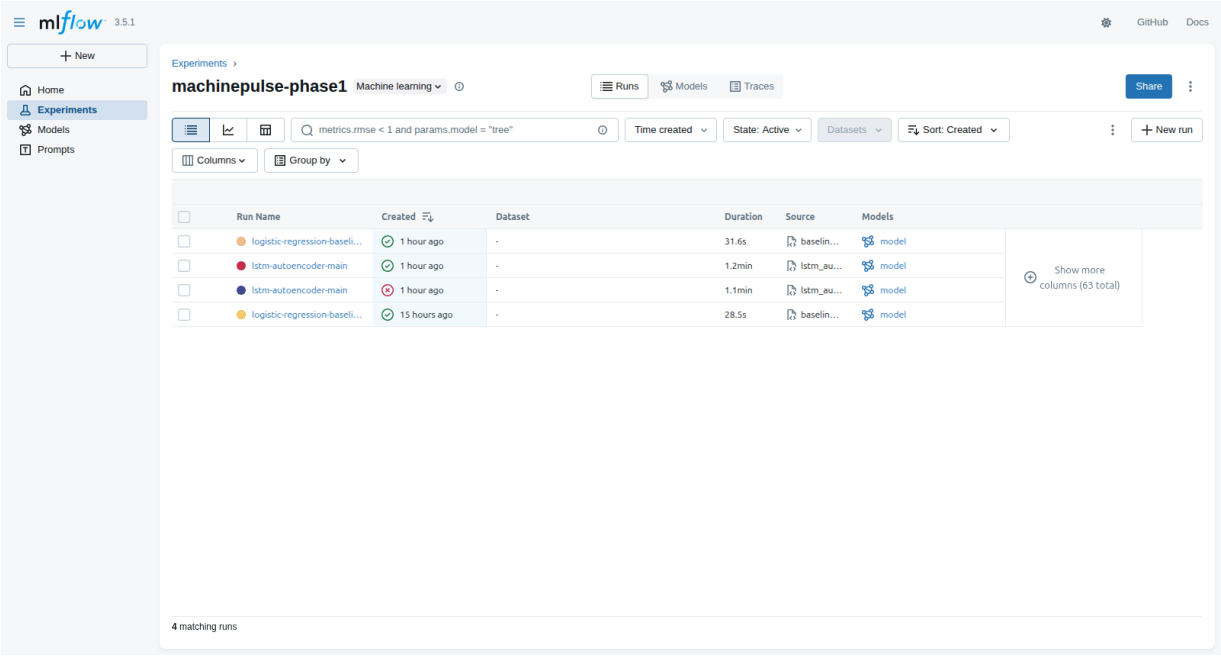


Figure 2. Remote machinepulse-phase1 experiment with multiple completed baseline and main-model runs; the red cross records the earlier failed registration attempt.

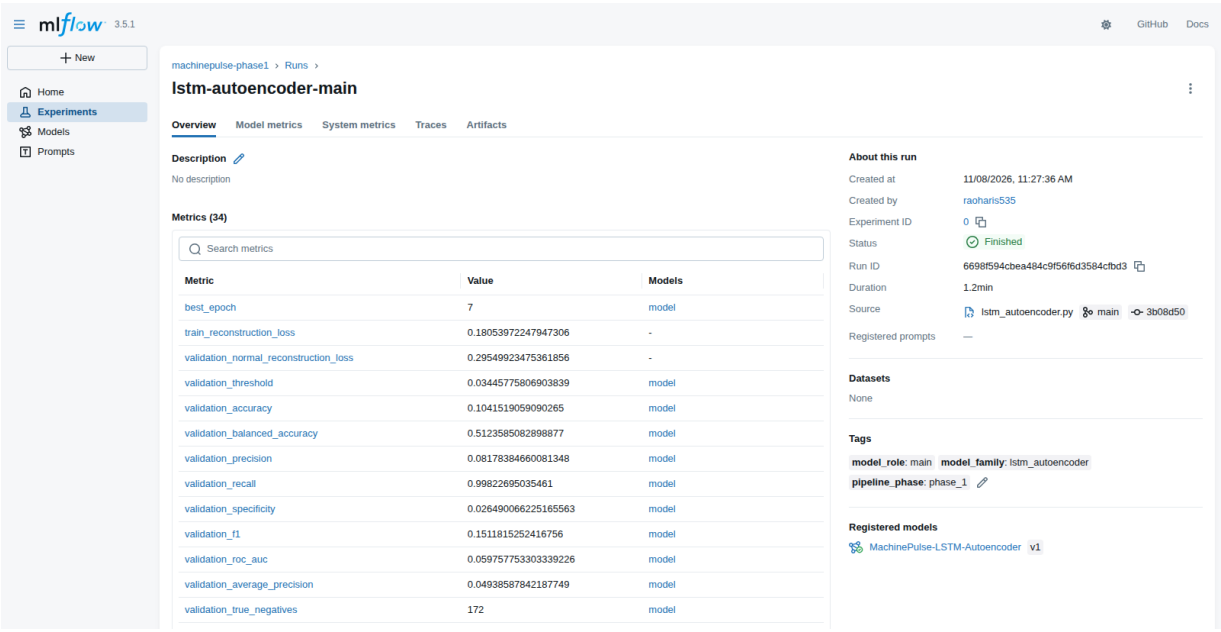


Figure 3. Successful LSTM run showing logged losses, validation metrics, hyperparameters, tags, Finished status, and the registered-model association.

Model registry and reproducibility

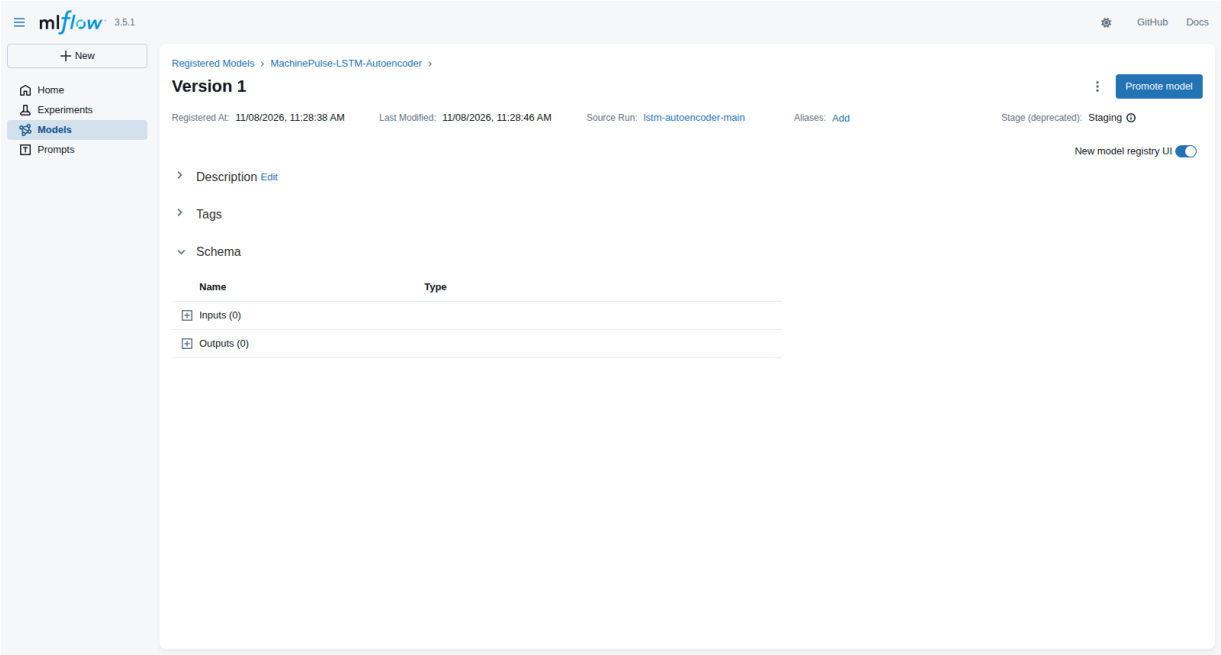


Figure 4. MachinePulse-LSTM-Autoencoder version 1 registered on the remote MLflow server with lifecycle stage Staging.

Repeatable execution

A fresh authorized clone can restore DVC outputs and reproduce the entire pipeline with the following commands. Remote credentials are stored only in the ignored .dvc/config.local file; raw data, model weights, .env files, and tokens are excluded from Git.

```
git clone https://github.com/RaoHaris535/MachinePulse-Project.git
cd MachinePulse-Project && uv sync --group dev
source .venv/bin/activate && dvc pull
dvc repro && pytest -q && ruff check .
dvc push && dvc status -c
```

Evaluation interpretation and limitation

The chronological split exposes severe incident-to-incident distribution shift. The baseline transfers poorly to the final incident. The LSTM ranks test anomalies well (ROC-AUC 0.9825; average precision 0.2787), but the validation-selected threshold predicts every test sequence as anomalous, producing low precision and F1. This is documented rather than hidden or corrected using test labels. Future work should add calibration across multiple historical failure periods and drift monitoring, which are outside Phase 1.

Conclusion

Phase 1 satisfies the required local MLOps foundation: source-controlled code and configuration, DVC-versioned artifacts on a remote, validation-gated processing, two evaluated systems, remote MLflow tracking, a Staging registered model, and an end-to-end named pipeline. The repository is ready for later deployment and monitoring phases without mixing those concerns into this submission.

Prepared 8 November 2026 | Dataset: MetroPT-3, UCI Machine Learning Repository (DOI 10.24432/C5VW3R)